Lightweight Hybrid CNN–Attention-Based Vision Transformer (LHC-AViT) for Hyper Spectral Image Classification

(925EC5006, Manda Sreevalli, Executive Phd)

1. Problem Statement

Hyperspectral Image (HSI) classification represents a computationally and theoretically demanding frontier in remote sensing, primarily due to the inherent complexity of high-dimensional spectral data. The core challenges involve the non-linear spectral mixing of hundreds of contiguous bands, significant spatial variability across heterogeneous landscapes, and the persistent Hughes Phenomenon, or "Curse of Dimensionality," exacerbated by the scarcity of high-quality labeled samples [1]. Successfully mapping these complex discriminative feature manifolds requires architectures capable of simultaneous local and global feature extraction without succumbing to over-parameterization or informational redundancy. Existing deep learning paradigms often suffer from specific architectural trade-offs. While Convolutional Neural Networks (CNNs) are adept at capturing hierarchical spatial patterns, their fixed receptive fields constrain their ability to model long-range spectral-spatial dependencies. Conversely, standard Vision Transformers (ViTs) excel at global modeling through self-attention but lack the inductive biases—specifically locality and translation equivariance—intrinsic to CNNs [2]. This gap often results in models that are either spatially sensitive but spectrally shallow, or globally aware but computationally prohibitive and data-inefficient. This motivates the development of a lightweight and dynamically adaptive hybrid architecture capable of efficient spectral–spatial modeling under real-world constraints.

2. Research Gaps

Despite advances in hybrid modeling, several critical structural and methodological voids persist in current HSI research:

Structural Gaps

- Dimensionality Reduction and Geometric Loss: Traditional 2D-based fusion and rasterization techniques often collapse 3D spectral-spatial cubes into 2D formats, preventing the model from learning direct 3D geometric features. This results in significant information loss for vertically defined semantic objects, such as distinguishing tree canopies from underlying ground surfaces in urban environments [3], [4].
- The Gating Gap: The implementation of learnable gating mechanisms remains an under-explored area. Most architectures lack the dynamic flexibility to balance spatial and spectral streams based on the sample-specific importance of a given pixel's neighborhood [2].

Methodological Gaps

- Inflexible Fusion Strategies: Conventional "naive" fusion methods, including simple concatenation or element-wise addition, are inherently static. These operations fail to resolve "modal dominance," where one feature stream overshadows another, leading to feature interference and suboptimal representations [4].
- Static vs. Dynamic Importance: Current strategies do not account for the varying relevance of spatial structures versus spectral signatures across different land-cover classes, neglecting the dynamic interdependencies required for robust spectral-spatial joint representation [2], [4].

3. Proposed Methodology

The proposed **LHC-AViT (Lightweight Hybrid CNN–Attention Vision Transformer)** architecture utilizes a dual-branch framework designed for efficient feature extraction and adaptive integration. Preprocessing begins with Principal Component Analysis (PCA) to reduce the spectral dimension to $C = 30$ components, thereby mitigating the Hughes Phenomenon while retaining over 99% of the spectral variance. From this reduced cube, 3D patches ($15 \times 15\ Pixels$) are extracted to serve as the local context for each labeled pixel.

3.1 Local Feature Attention Module (LFAM)

The LFAM serves as the spatial extraction branch, employing a sequence of 2D convolutional blocks to learn hierarchical geometric patterns. The module consists of two stages: the first utilizes 32 filters and the second utilizes 64 filters, both using (3×3) kernels. Each stage incorporates Batch Normalization (BN) and ReLU activation to ensure stable training and non-linearity. The branch culminates in an adaptive average pooling layer, which compresses the hierarchical features into a localized spatial descriptor, $f_{CNN} \epsilon R^D$.

3.2 Multimodal Interaction Modeling (MIM)

The spectral branch utilizes a Transformer encoder to explicitly model long-range dependencies. The input patch is reshaped into a sequence of tokens, linearly projected into a $D -$ dimensional embedding space, and combined with learnable positional encodings. To address the gap in multimodal interaction, this module incorporates a bidirectional weighting concept inspired by CrossPointAttention (CPA). In this stage, the spectral branch "attends" to spatial features to resolve potential modal dominance, ensuring that the spectral signature of the central pixel is refined by its geometric context [4]. The Transformer encoder employs a linear-complexity attention mechanism (e.g., kernel-based attention), reducing computational complexity from $O(N^2)$ to $O(N)$. The resulting features are then aggregated using global average pooling to produce the spectral representation, $f_{ViT} \epsilon R^D$.

3.3 Adaptive Fusion Module (AFM)

The AFM integrates the spatial descriptor f_{CNN} and the spectral representation f_{ViT} through a data-driven gating mechanism. Both vectors are projected to a common embedding dimension D (typically 64 or 128) to facilitate alignment. The vectors are concatenated and processed by a Gating MLP with a Sigmoid activation to generate a scalar gate value $G \epsilon [0,1]$. The final fused feature vector, f_{fused}, is derived using the adaptive weighting formula:

$$f_{fused} = G.f_{CNN} + (1 - G).f_{ViT}$$

This allows the model to dynamically emphasize the most discriminative modality on a per-sample basis. The system is optimized using categorical cross-entropy loss:

$$L_{CE} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{k=1}^{K} y_{i,k} \log(\hat{y}_i, k)$$

Unlike existing hybrid models, the proposed framework introduces a sample-adaptive fusion mechanism combined with cross-modal interaction refinement, enabling dynamic spectral–spatial feature integration and improved robustness under limited training conditions.

4. Formal Research Objectives

1. Develop a hybrid architecture that integrates the inductive locality of CNNs with the global spectral modeling capabilities of ViTs to overcome individual structural limitations.
2. Implement a lightweight, attention-based gating mechanism (AFM) that dynamically assigns sample-specific importance to spatial and spectral feature streams, mitigating feature interference.
3. Achieve state-of-the-art (SOTA) performance across Indian Pines, Pavia University, and Salinas Valley benchmark datasets, prioritizing Overall Accuracy (OA) and Kappa coefficients.
4. Validate computational efficiency by maintaining rapid training times, specifically targeting the 114.97-second benchmark established for the Salinas Valley dataset [2], ensuring viability for real-time remote sensing.
5. Validate robustness against noise by generating high-coherence classification maps that exhibit a measurable reduction in "salt-and-pepper" noise and improved class boundary definition.

5. Expected Contributions

The proposed model achieves a balance between accuracy, computational efficiency, and data efficiency, addressing key limitations of existing HSI classification models.

- Algorithmic Innovation: The introduction of a data-driven, adaptive weighting mechanism that replaces static fusion with a learnable gate, allowing for context-aware feature integration.
- Performance Benchmarks: Anticipated achievement of SOTA accuracy, specifically an OA exceeding 98.6% on Indian Pines and 99.9% on Pavia University [2].
- Computational Efficiency: Advancement of lightweight strategies to make hybrid models viable for deployment, providing a baseline for future work in model pruning and knowledge distillation.
- Visualization and Interpretability: Production of classification maps with superior discriminative capability, clearly delineating classes with ambiguous geometric structures such as bare earth and paved parking lots compared to SST-FA and GroupFormer methods.

6. References

[1] A. Adityayuda, A. A. Supriyadi, and S. Arief, "Development of a remote sensing system for real-time detection of military threats," in *2024 IEEE Asia-Pacific Conference on Geoscience, Electronics and Remote Sensing Technology (AGERS)* , 2024, pp. 257–268.

[2] G. Farooque, Q. Liu, A. B. Sargano, and L. Xiao, "Swin transformer with multiscale 3d atrous convolution for hyperspectral image classification," *Eng. App. of AI* , vol. 126, 2023.

[3] P. Ghamisi, B. Höfle, and X. X. Zhu, "Hyperspectral and LiDAR data fusion using extinction profiles and deep convolutional neural network," *IEEE J-STARS* , vol. 10, no. 6, pp. 3011–3024, 2017.

[4] A. Rizaldy, R. Gloaguen, F. E. Fassnacht, and P. Ghamisi, "HyperPointFormer: Multimodal Fusion in 3-D Space With Dual-Branch Cross-Attention Transformers," *IEEE J-STARS* , 2025.